

PHARMACODYNAMICS

(1st year foundation module)

By

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DIKIOHS/DUHS

LEARNING OUTCOMES

PHARMACODYNAMICS

1) Mechanism of action: Receptors: Types and Actions, Signaling mechanisms

To define receptor

To determine different types of receptors

To describe different signaling mechanism.

2) Drug Dose relationship

To define dose-response relationship

To classify different dose response relationship curve

To Define potency, efficacy, ED₅₀, LD₅₀, Therapeutic index

3) Agonist/Antagonist

To define agonist, full agonist, partial agonist, inverse agonist

To define functional antagonist and pharmacological antagonist

To compare competitive and noncompetitive antagonism

To describe addition, synergism, potentiation and pharmacological antagonism)

PHARMACODYNAMICS

- ▣ **Actions Of Drug On The Body**
- ▣ **Mechanism Of Action (MOD)**
- ▣ **Influence Of Drug Concentration On Magnitude Of Response**

RECEPTORS

- ❑ **SPECIALIZED TARGET
MACROMOLECULES**
- ❑ **MOST ARE PROTEINS**
- ❑ **CELL SURFACE, CYTOPLASM OR IN
THE NUCLEUS**

FUNCTION OF RECEPTOR

- Recognition Of Ligand (Drug i.e Agonist)**
- Act As Signal Detectors**
- Transduction Into Response**

RECEPTOR DOMAINS

- LIGAND BINDING DOMAIN**
- EFFECTORS DOMAIN (initiate 2nd messenger response)**

“ACTION”:

- CONFORMATIONAL CHANGE (AGONIST/Response)**
- PREVENTION OF CONFORMATIONAL CHANGE (ANTAGONIST/ No response)**

“EFFECT”:

- CHANGE IN BIOLOGICAL FUNCTION**

DRUG AFFINITY

- DRUG CAPACITY TO BIND TO RECEPTOR
- TO FORM THE COMPLEX WITH ITS RECEPTOR
(Drug Receptor COMPLEX)

INTRINSIC ACTIVITY

- TRIGGER PHARMACOLOGICAL RESPONSE (beta 1 receptors in heart causes tachycardia and M2 receptors in the heart causes bradycardia)

LIGAND

- MOLECULE ATTACHES SELECTIVELY TO RECEPTORS OR OTHER SITES
(epinephrine/norepinephrine/acetylcholine)

SIGNAL TRANSDUCTION

**DRUG-RECEPTOR COMPLEX INITIATES
ALTERATIONS IN BIOCHEMICAL OR
MOLECULAR ACTIVITY OF A CELL**

Receptor States

Inactive (R)

Active (R₊)

AGONIST: Increase fraction of R₊

PARTIAL AGONIST: Increase Fraction of R₊ but less than Agonist

ANTAGONIST: Increase fraction of R

MECHANISM OF DRUG ACTION

TARGET SITES OF DRUGS

- **ENZYMES**
- **ION CHANNELS**
- **TRANSPORTERS**
- **RECEPTORS**

MECHANISM OF ACTION

- **Some Drugs Act By Physical Or Chemical Properties**

Examples :

Antacids neutralize gastric acid
Mannitol acts as osmotic diuretic

- **Majority Act By Binding To Specific Receptors**

NATURE OF RECEPTORS

Majority are protein in nature

enzymatic protein

Dihydrofolate reductase

e.g.

TRIMETHOPRIM

Monoamine oxidase

e.g.

PHENELZINE

Structural protein:

Tubulin

e.g **COLCHICINE**

Transport protein:

Na-K-ATPase

e.g.

DIGITALIS.

H-K-ATPase

e.g.

OMEPRAZOLE

Ionic channel:

Calcium channels

e.g. Ca^{++} **BLOCKERS**

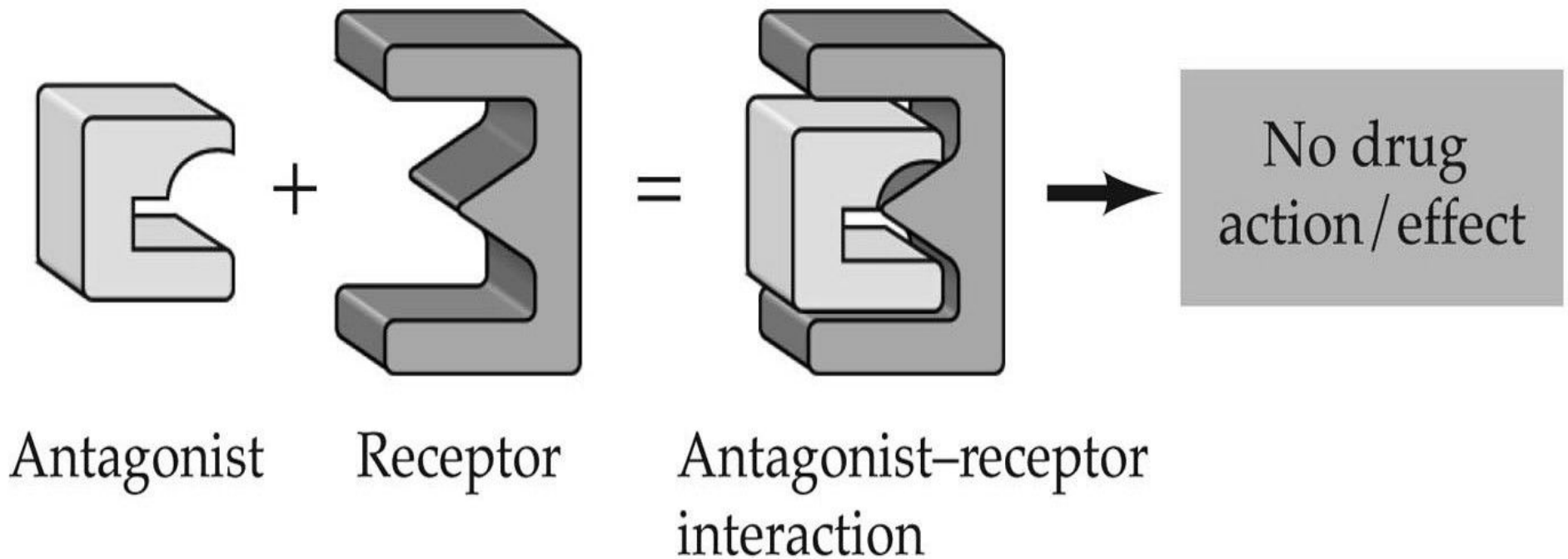
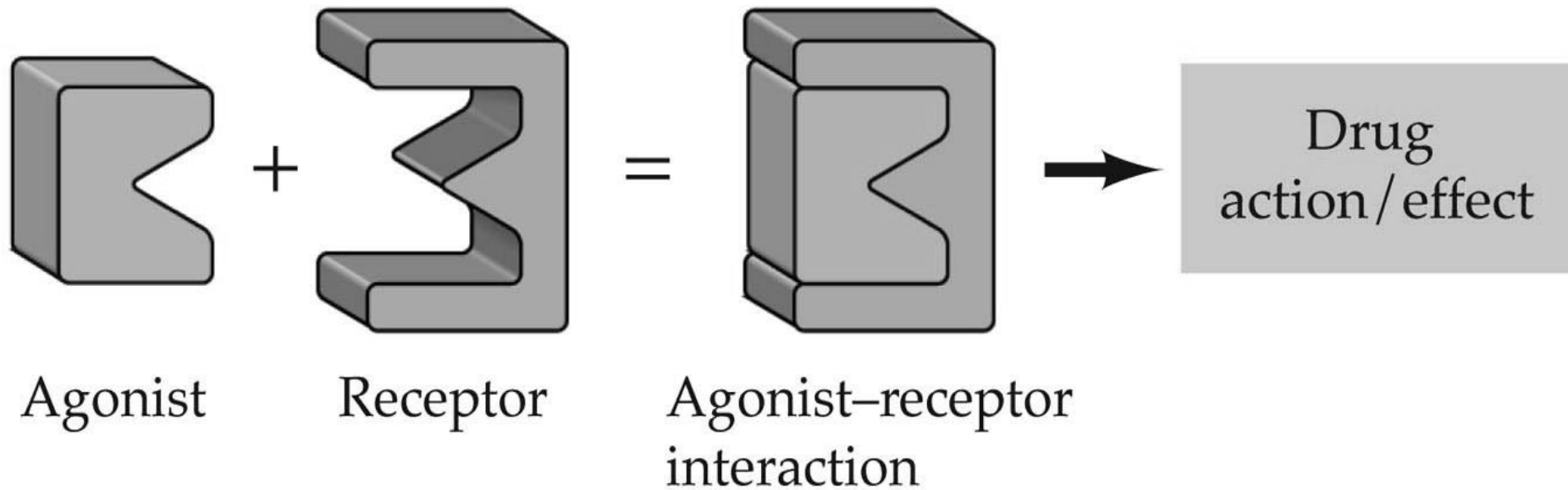
Sodium channels

e.g.

ANESTHETICS

Nucleic acids (RNA & DNA):

For antiviral and anticancer drugs



MOLECULAR MECHANISM OF DRUG ACTION

TRANSMEMBRANE SIGNALING

- 1. INTRACELLULAR RECEPTORS FOR LIPID SOLUBLE AGENTS (GENE ACTIVE RECEPTORS)**
- 2. LIGAND GATED TRANSMEMBRANE ENZYME**
- 3. LIGAND AND VOLTAGE GATED ION CHANNEL**
- 4. G-PROTEIN COUPLED RECEPTORS**

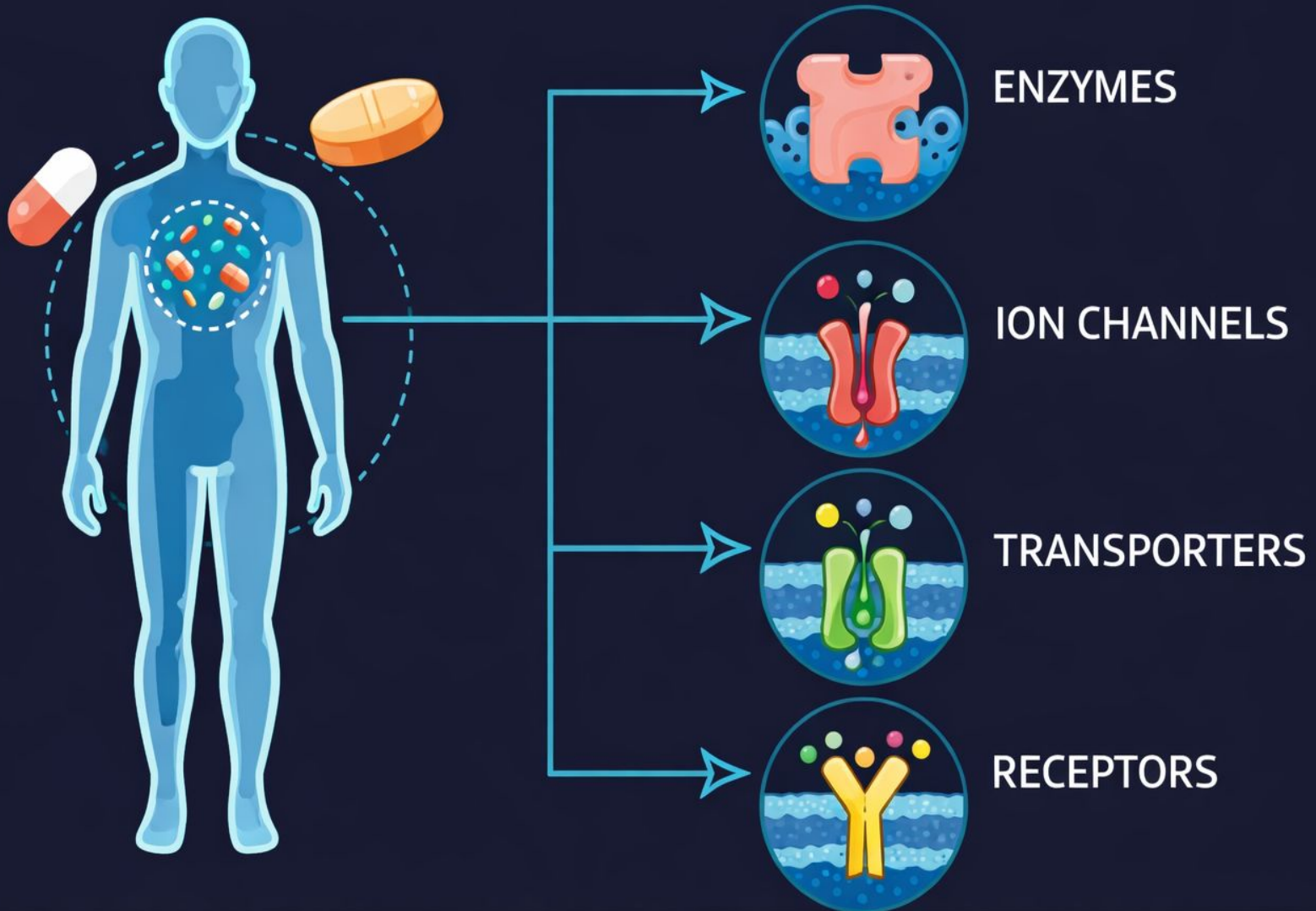
NON-RECEPTOR MEDIATED DRUG ACTION

EXAMPLES

- **Physical And Chemical Means –
E.G Antacids, Cholestyramine**
- **Alkylating Agents-binding With Nucleic Acid
E.G. Mechlorethamine**
- **Antimetabolites: Purine And Pyrimidine
Analogues –
E.G. 6 Mp And 5 Fu**

PHARMACODYNAMICS

TARGET SITES OF DRUGS



Sites of Drug Action



TYPES OF RECEPTORS

Type I: Ionotropic receptors

(Transmembrane Ligand gated ion channels)

Type II: Metabotropic receptors

(G proteins coupled receptors(GPCR))

1. Adenyl cyclase: cAMP system
2. Phospholipase-C: Inositol Phosphate system
3. Ion channel regulation

TYPES OF RECEPTORS & SIGNAL TRASDUCTION

Type III: Enzyme linked receptors

Tyrosine Kinase

Type IV: Intracellular Receptors

(Receptors regulating gene expression)

A Ligand-gated ion channels

Example:

Cholinergic nicotinic receptors

B G protein-coupled receptors

Example:

α and β adrenoreceptors

C Enzyme-linked receptors

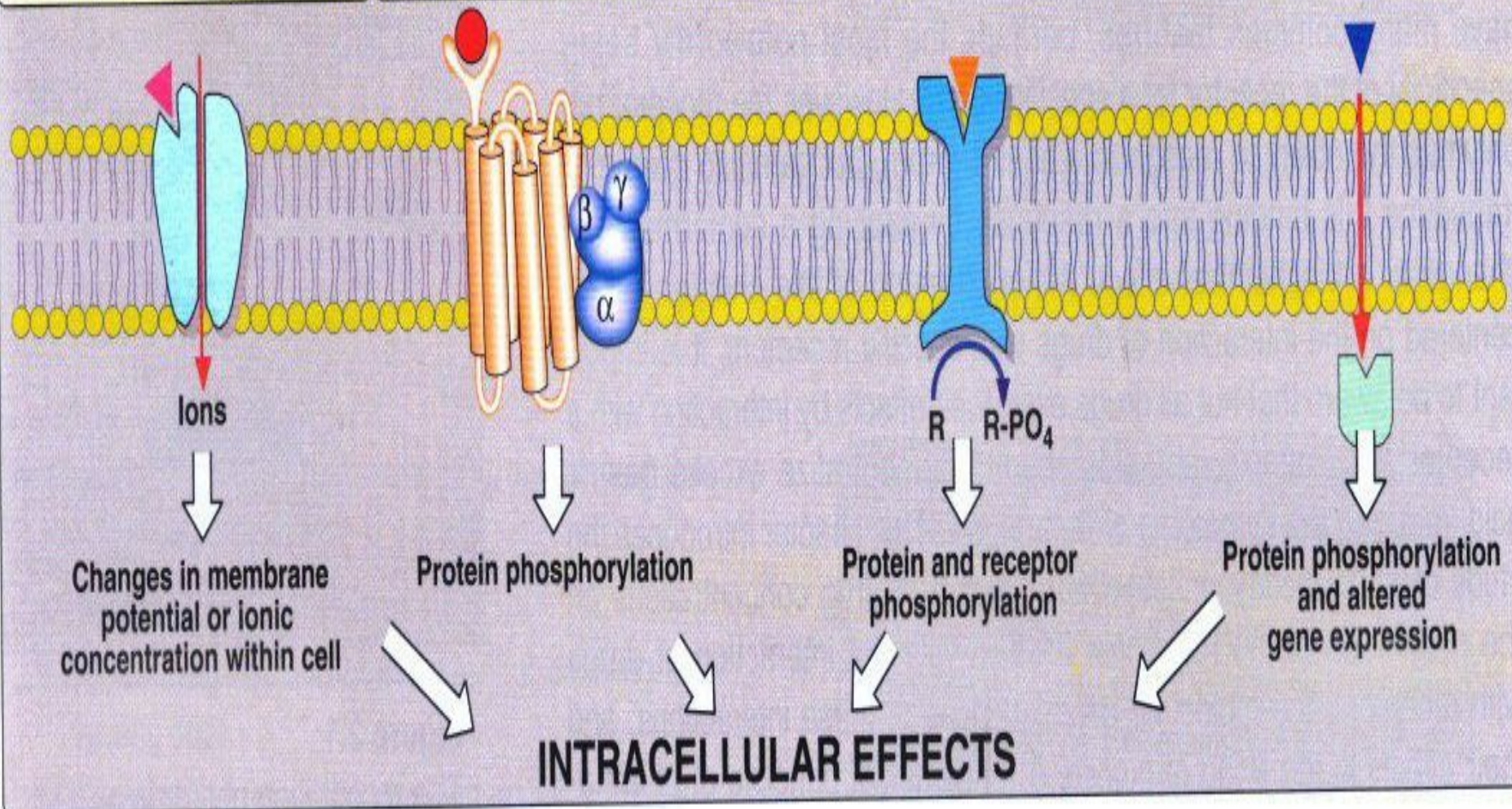
Example:

Insulin receptors

D Intracellular receptors

Example:

Steroid receptors



LIGAND AND VOLTAGE GATED ION CHANNELS

LIGAND GATED

- ❑ LIGAND GATED BINDING SITE EXTRACELUULARLY
- ❑ ON ACTIVATION OF RECEPTOR BY AGONIST ION CHANNELS OPEN
- ❑ FUNCTION: NEUROTRANSMISSION, CARDIAC OR MUSCLE CONTRACTION INCREASE TRANSMEMBRANE CONDUCTANCE
- ❑ E.G NICOTINIC-ACETYLCHOLINE, GABA RECEPTOR STIMULATION

VOLTAGE GATED

- ❑ DO NOT BIND NEUROTRANSMITTERS
- ❑ DIRECTLY INHIBIT ION CHANNEL
- ❑ E.G LOCAL ANESTHETICS ON SODIUM CHANNEL

G-PROTEIN COUPLED RECEPTORS

- CELL MEMBRANE RECEPTORS
- EXCTRACELLULAR DOMAIN (FOR AGONIST) AND INTRACELLULAR DOMAIN(G-PROTEIN OR EFFECTOR MOLECULE)
- 7 A-HELICAL
- HAVE ALFA, BETA AND GAMMA SUBUNITS

TYPES OF G-PROTEINS

G protein Effectors pathway Substrates

1) G_s----- Adenylyl cyclase - ----Beta-receptors, H₂,

2) G_i----- Adenylyl cyclase -----M₂

D₂, alpha-2

3) G_q----- PLC - IP₃/DAG----- Alpha-1, H₁, M₁,M₃

GPCR - 3 MAJOR PATHWAYS

**1. ADENYLYL CYCLASE: CAMP
PATHWAY**

**2. PHOSPHOLIPASE C: IP₃-DAG
PATHWAY**

ENZYME LINKED RECEPTORS

**CONSIST OF PROTEIN THAT FORM COMPLEXES
WHEN ACTIVATED UNDERGO CONFORMATIONAL
CHANGES RESULTING IN INCREASED ENZYME
ACTIVITY**

**E.G. TYROSINE KINASE OR
GUANYL CYCLASE DOMAIN**

LIGAND REGULATED TRANSMEMBRANE ENZYME

- ❑ EC HORMONE BINDING DOMAIN
- ❑ CYTOPLASMIC ENZYME DOMAIN
- E.G TYROSINE KINASE, GUANALYL CYCLASE
- ❑ LIGAND BINDING TO ECD
- ❑ CONFORMATIONAL CHANGE
- ❑ ACTIVATION OF TK
- ❑ PHOSPHORYLATION OF TYROSINE RESIDUES
- ❑ E.G. INSULIN,EGF,PDGF,ANP, TGF BETA

INTRACELLULAR RECEPTORS FOR LIPID SOLUBLE AGENTS

- LIPID SOLUBLE LIGAND**
- EFFICIENTLY CROSS LIPID BILAYER**
- ACT ON INTRACELLULAR RECEPTORS**
- REGULATING GENE EXPRESSION**

- E.G STEROIDS,VIT D, THYROID HORMONE**
- OTHER EXAMPLES**
- PACLITAXIL, TRIMETHOPRIM, ERYHTROMYCIN**

RECEPTOR DOWN REGULATION (Desensitization)

- ❑ DECREASE EXPRESSION OF RECEPTORS**
- ❑ CONTINUED EXPOSURE TO AN AGONIST**
- ❑ INTERNALIZATION OF THE RECEPTORS**
- ❑ DECREASED SYNTHESIS**
- ❑ INCREASED DESTRUCTION**
- ❑ E.G. MORPHINE**

RECEPTOR UPREGULATION (Sensitization)

- ❑ INCREASE EXPRESSION OF RECEPTORS**
- ❑ PROLONGED DEPRIVATION OF AGONIST**
- ❑ SUPERSENSITIVITY OF THE RECEPTOR**
- ❑ E.G. PROPRANOLOL, CLONIDINE**

SPARE RECEPTORS

when a drug produces a maximum effect by binding to a less number of available receptors; the receptors remaining are spare receptors

ORPHAN RECEPTORS

Natural ligands are presently unknown

ORPHAN DRUGS

drugs for rare diseases

- ❑ **DOSE RESPONSE CURVE**

- ❑ **POTENCY**

- ❑ **EFFEICACY**

- ❑ **ED₅₀**

- ❑ **LD₅₀**

- ❑ **THERAPEUTIC INDEX**



DOSE:

Quantity Of A Drug Given To A Patient

At One Time To Produce A Certain

Therapeutic Effect.

DRUG + RECEPTOR



RESPONSE

DOSE RESPONSE RELATIONSHIP

**RELATIONSHIP BETWEEN
THE DOSE OF THE DRUG
AND THE MAGNITUDE OF
THE RESPONSE**

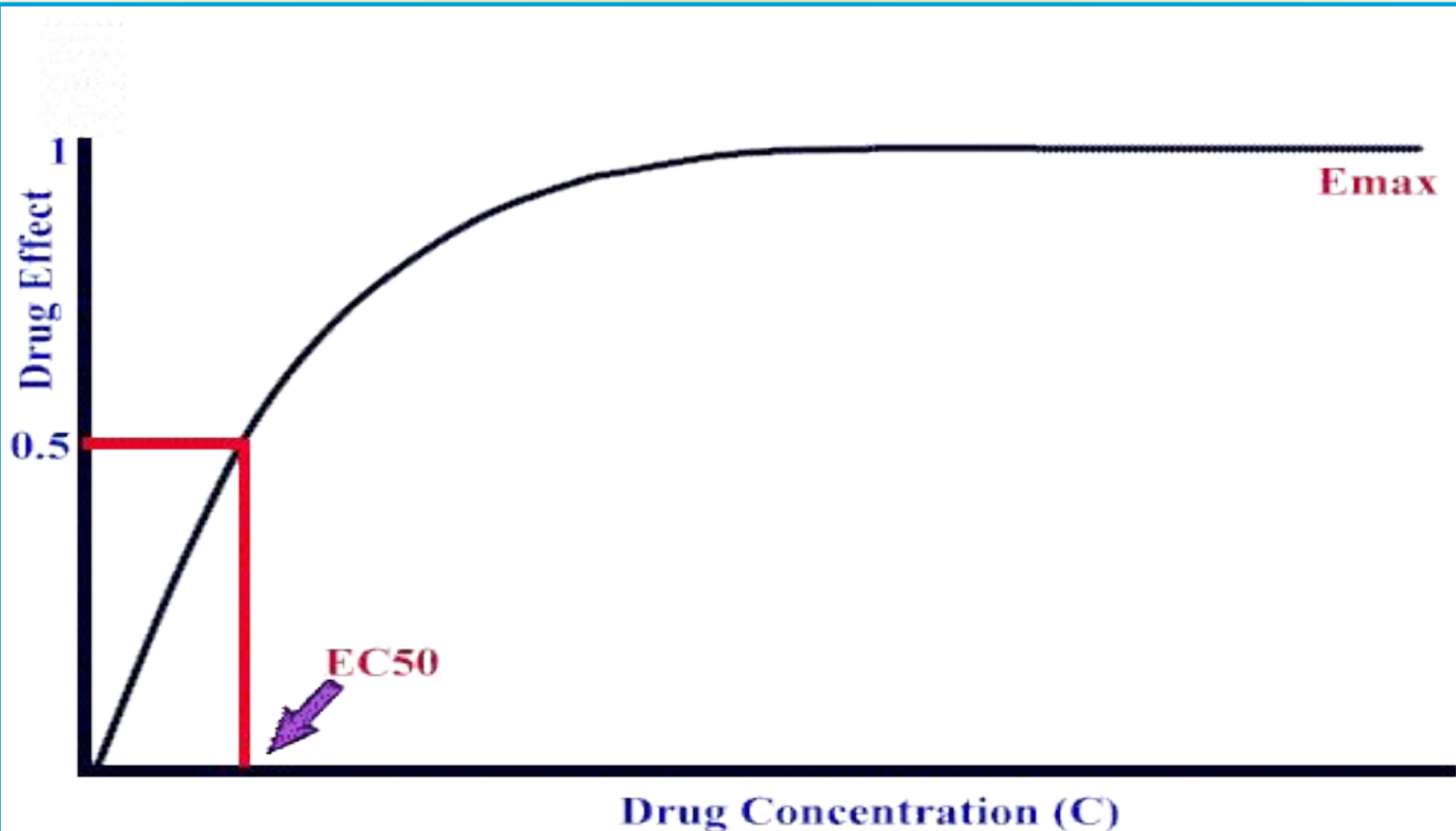
Shape of the curve

- A standard dose-response curve is defined by four parameters:
- the baseline response (Bottom),
- the maximum response (Top),
- the slope, and the
- drug concentration that provokes a response halfway between baseline and maximum (EC50).

Dose-Response curve

DOSE-RESPONSE
RELATIONSHIP

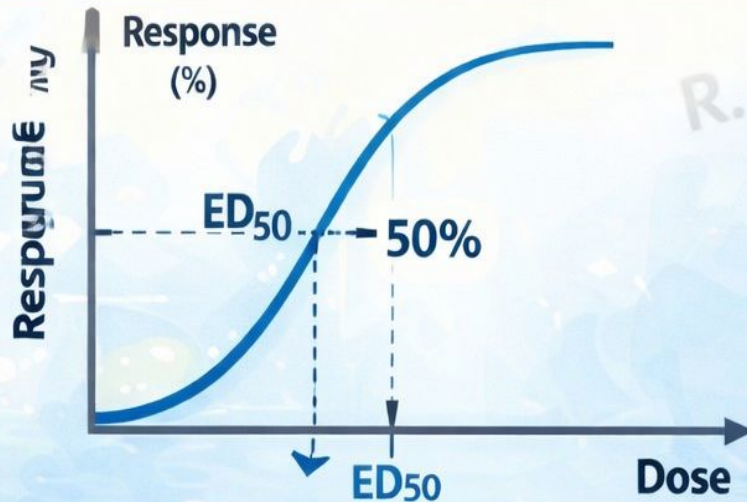
- Simple X-Y graph
- Magnitude of response to the dose



Dose-Response Relationship Curves

Graded

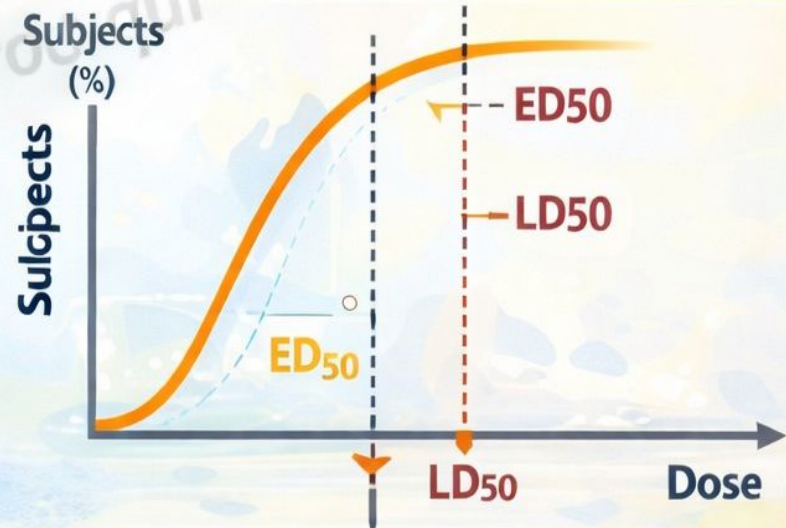
Graded Dose-Response Curve



Varies by Degree

Quantal

Quantal Dose-Response Curve



Varies by Presence / Absence

$$\text{Therapeutic Index} = \frac{\text{LD}_{50}}{\text{ED}_{50}}$$

$$\frac{\text{ED}_{00}}{\text{Effective Dose}} \longleftrightarrow \frac{\text{LD}_{50}}{\text{Lethal Dose}}$$

A. GRADED DOSE-RESPONSE RELATIONS

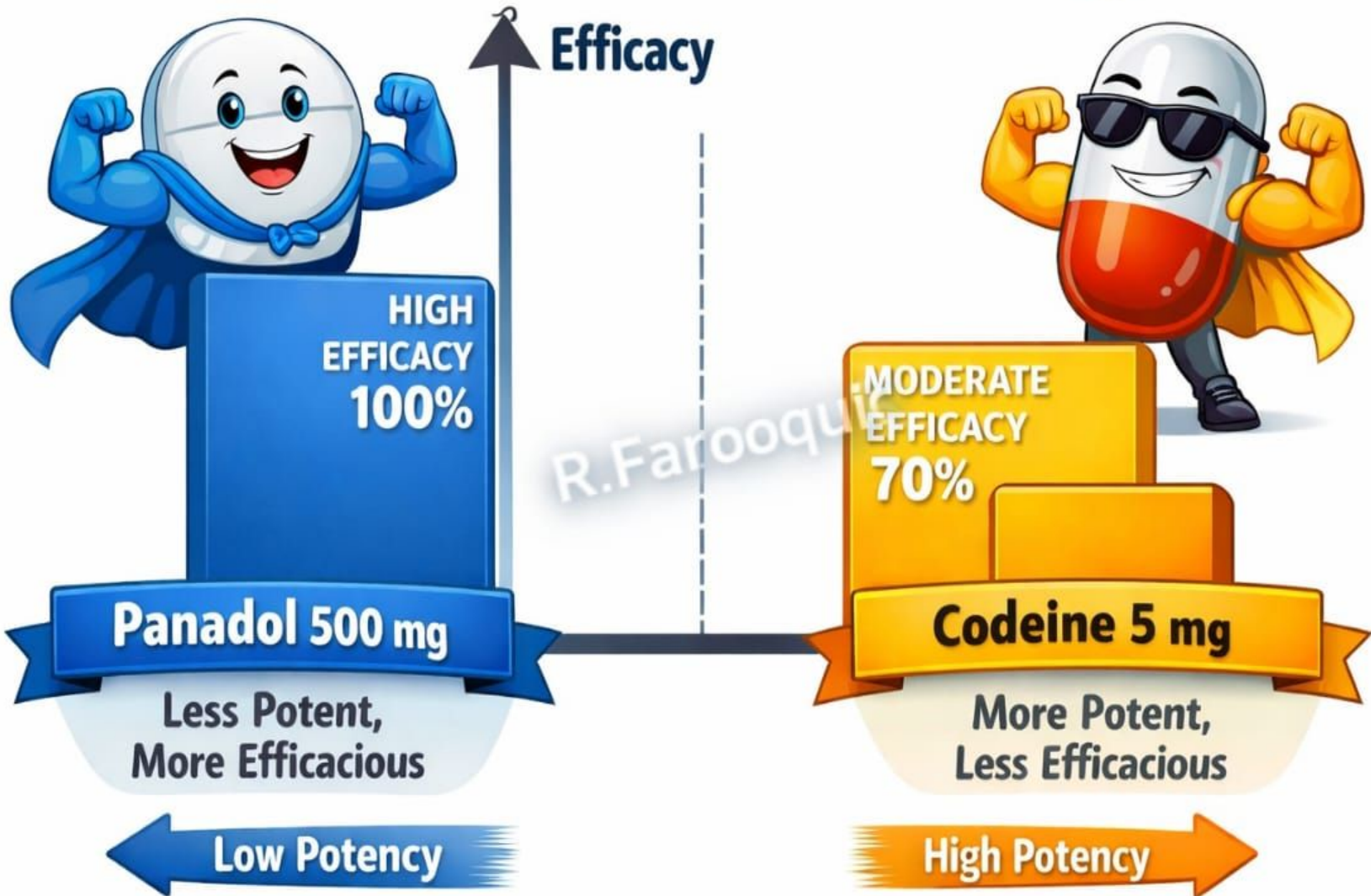
Potency

- Range of doses
- Produce a response

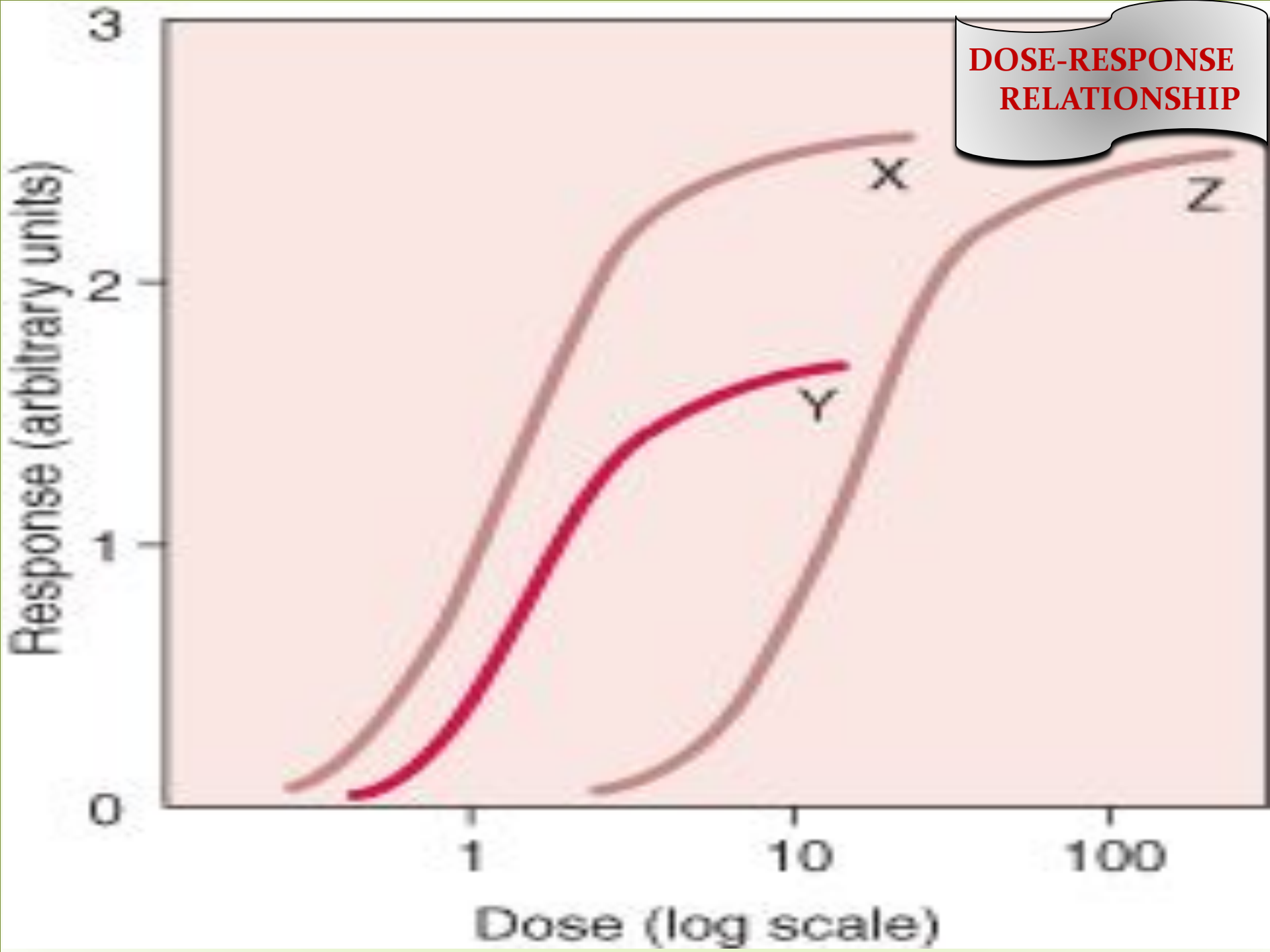
Maximal efficacy

- Maximal response produced

Potency vs Efficacy



**DOSE-RESPONSE
RELATIONSHIP**



B. QUANTAL DOSE-EFFECT CURVES

- Concentration of the drug**
- Percentage of a specific population**

Median effective dose (ED_{50})

□ **Effect 50% of individuals**

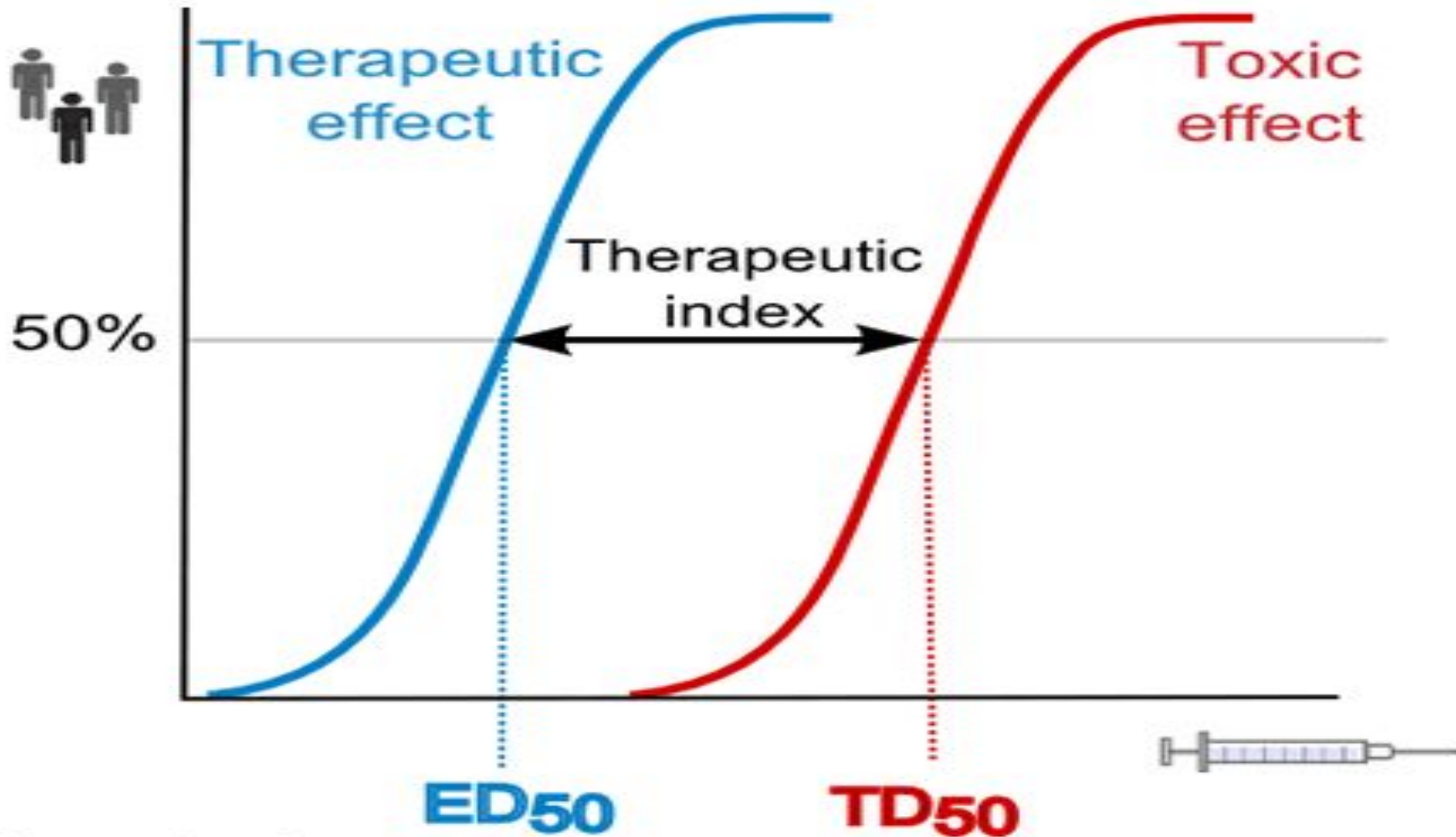
Median toxic dose (TD_{50})

□ **Toxic effect in 50% of
animals/human**

Median lethal dose (LD_{50})

□ **Death in 50% of animals**

DOSE-RESPONSE RELATIONSHIP



WTI

- ☐ Penicillin
- ☐ Aspirin
- ☐ Ibuprofen
- ☐ Diazepam

NTI

- ☐ Digoxin
- ☐ Amphotericin B
- ☐ Warfarin
- ☐ Phenytoin

WARFARIN

Narrow Therapeutic Index

Bleeding!



- Hemorrhage
- Blood in Urine

I have a **LOW THERAPEUTIC INDEX!**



Brain Bleed!



Intracranial Hemorrhage

Easy Bruising



- Severe Bruises

GI Bleeding!



Stomach & Intestinal Bleeds



Risk of Overdose!

Frequent Monitoring Needed!

Full Agonist



Full Agonist

Morphine

Partial Agonist



Partial Agonist

Buprenorphine

Inverse Agonist



Inverse Agonist

Naloxone

AGONIST

AGONIST

- ❑ ACTIVATES A RECEPTOR
- ❑ PRODUCE AN EFFECT
- ❑ SIMILAR TO A THAT OF THE PHYSIOLOGICAL SIGNAL MOLECULE
- ❑ E.G. MUSCARINE, NICOTINE

TYPES:

FULL AGONIST

- ❑ ACTIVATES A RECEPTOR
- ❑ PRODUCE AN EFFECT SIMILAR TO PHYSIOLOGIC S. MOLECULE
- ❑ HAVE FULL AFFINITY
- ❑ HIGH INTRINSIC ACTIVITY
- ❑ TRIGGER MAXIMAL BIOLOGICAL RESPONSE (E.MAX 100%)
- ❑ E.G. MORPHINE, EPINEPHRINE

AGONIST

PARTIAL AGONIST

- ❑ ACTIVATES A RECEPTOR
- ❑ PRODUCE RESPONSE
- ❑ HAVE FULL AFFINITY
- ❑ LOW INTRINSIC ACTIVITY (SUBMAXIMAL RESPONSE)
- ❑ PARTLY AS EFFECTIVE AS AGONISTS
- ❑ E.G. BUPRENORPHINE, CODEIN

INVERSE AGONIST

- ❑ ACTIVATES A RECEPTOR
- ❑ PRODUCE AN EFFECT
- ❑ OPPOSITE DIRECTION TO THAT OF THE AGONIST
- ❑ HAVE FULL AFFINITY
- ❑ INTRINSIC ACTIVITY RANGES BETWEEN 0 TO -1
- E.G. BETA CARBOLINE

ANTAGONIST

ANTAGONIST

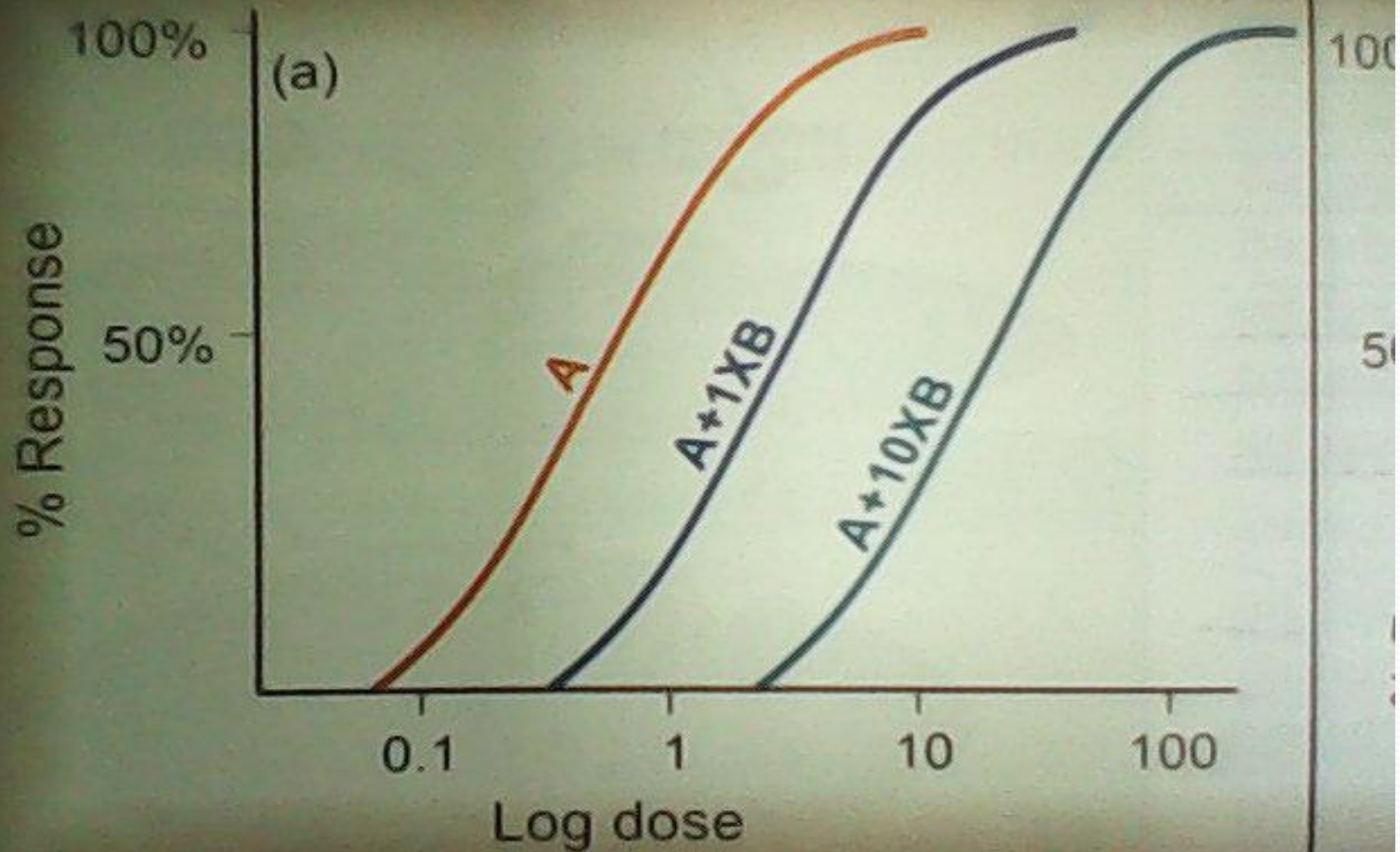
- ❑ PREVENTS THE ACTION OF AN AGONIST
- ❑ HAS AFFINITY
- ❑ NO INTRINSIC ACTIVITY
- ❑ DOES NOT HAVE AN EFFECT OF ITS OWN
- E.G. ATROPINE , PROPRANOLOL

ANTAGONIST

COMPETITIVE ANTAGONIST

- ❑ CHEMICALLY SIMILAR TO AGONIST
- ❑ COMPETES WITH AGONIST
- ❑ BINDS TO THE SAME SITE
- ❑ HAS AFFINITY
- ❑ NO INTRINSIC ACTIVITY
- ❑ NO RESPONSE
- ❑ SHORT DURATION OF ACTION
- ❑ LOG DRC OF THE AGONIST IS SHIFTED TO RIGHT
- ❑ HIGHER CONC. OF AGONIST OVERCOMES THE BLOCK (SURMOUNTABLE)
- ❑ AGONIST MAXIMUMAL EFFECT $EMAX_{100\%}$
- ❑ E.G. ATROPINE (antagonist) and Acetylcholine (agonist); MORPHINE (agonist) and NALOXONE (Antagonist)

Competitive Antagonism

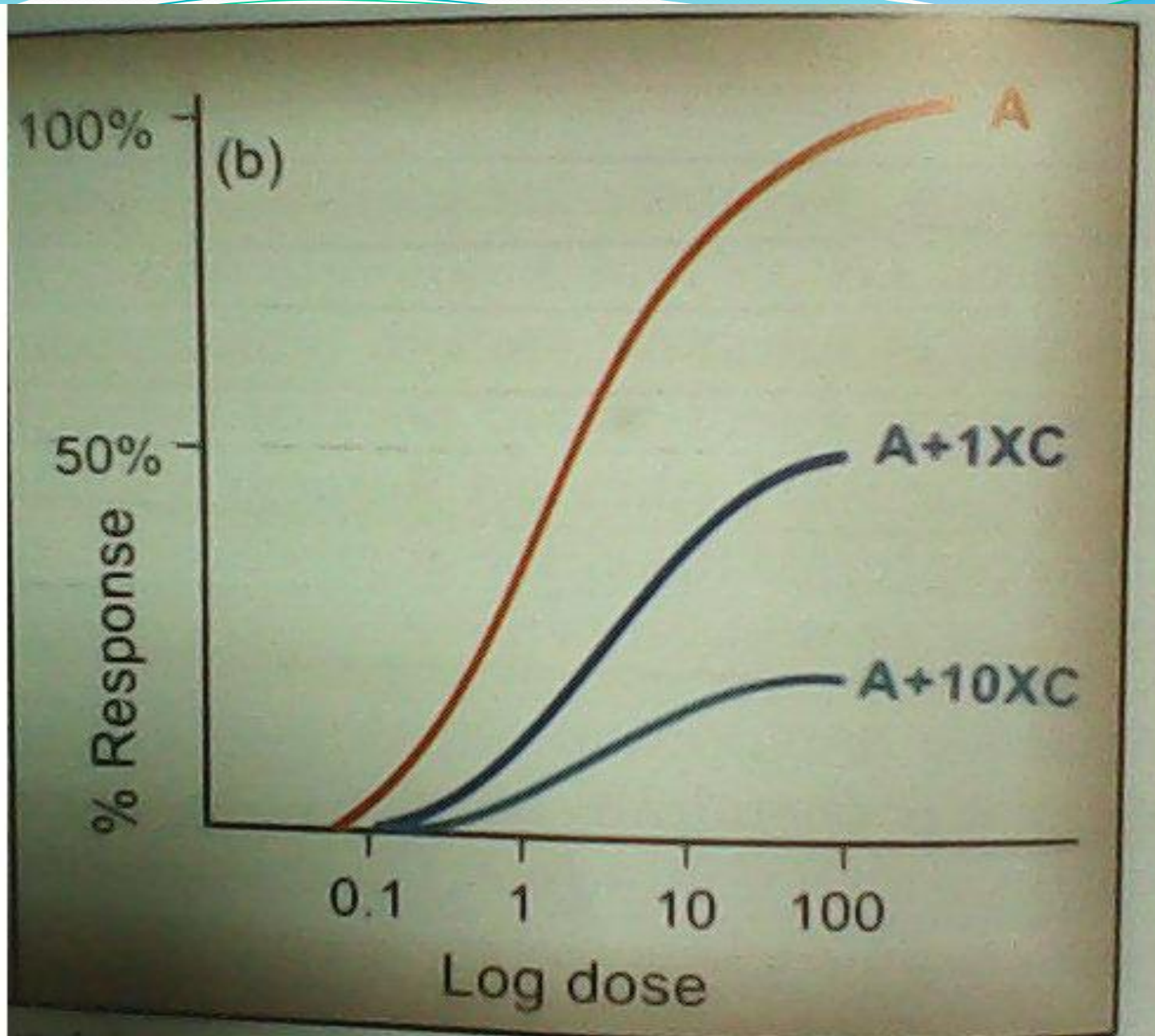


ANTAGONIST

NONCOMPETITIVE ANTAGONIST

- AGONIST -ANTAGONIST INTERACTION
NONCOMPETITIVE
- ANTAGONIST-RECEPTOR INTERACTION
IRREVERSIBLE
- AGONIST MAXIMAL RESPONSE IS LESS
- HIGH LEVEL OF AGONIST CONC.IS UNABLE TO
REVERSE THE BLOCK
- FLATTEN AGONIST DRC
- LONGER DURATION OF ACTION
- EXAMPLE: Phenoxybenzamine on α receptors.: E

Non-Competitive Antagonism



COMPARISON OF DIFFERENT TYPES OF ANTAGONISM

<u>COMPETITIVE REVERSIBLE</u>	<u>COMPETITIVE IRREVERSIBLE</u>	<u>NON-COMPETITIVE</u>
• Both agonist & antagonist bind to same receptor site. • Binding is weak and reversible. • Antagonist structurally resembles agonist	• Both agonist & antagonist bind to same receptor site. • Binding is strong (covalent) & irreversible • Antagonist structurally resembles agonist	• Agonist & Antagonist bind to different sites on same receptor • Binding is strong & usually irreversible • Antagonist does not structurally resemble agonist

Types of Antagonism

Pharmacological

Antagonist Blocker



Antagonist Blocks Agonist

Chemical Antagonism



Neutralization

Chemical Neutralizes Another Drug

Physiological Antagonism



Drug A
(Albuterol)

Drug B
(Histamine)

One Drug Opposes Another's Effects

Low Potency

Neutralization

Opposing Effects

MECHANISM OF ANTAGONISM

ANTAGONISM:

ONE DRUG DECREASES OR ABOLISH THE ACTION OF ANOTHER DRUG

- 1) **CHEMICAL**
- 2) **PHARMACOLOGICAL**
 - A) **COMPETITIVE ANTAGONISM**
 - B) **NON COMPETITIVE ANTAGONISM**
- 3) **PHYSIOLOGICAL**

MECHANISM OF ANTAGONISM

CHEMICAL ANTAGONISM

- ❑ TWO DRUGS REACT CHEMICALLY
- ❑ FORM AN INACTIVE PRODUCT
- ❑ WITHOUT INVOLVEMENT OF DRUG RECEPTORS
- ❑ E.G.
 - KMnO_4 oxidizes Alkaloid
 - Heparin and Protamine sulfate

MECHANISM OF ANTAGONISM

PHYSIOLOGICAL/FUNCTIONAL ANTAGONISM

- ❑ TWO DRUGS
 - ❑ ACT ON DIFFERENT RECEPTORS
 - ❑ BY DIFFERENT MECHANISMS
 - ❑ HAVE OPPOSITE EFFECTS
 - ❑ ON THE SAME PHYSIOLOGICAL FUNCTION
-
- GLUCAGON & INSULIN
 - **Histamine** and **Epinephrine** on bronchial smooth muscles.
 - **Acetylcholine** and **Epinephrine** on conducting system of the heart

MECHANISM OF ANTAGONISM

PHARMACOLOGICAL ANTAGONISM

ONE DRUG BLOCK (ANTAGONIST) THE RECEPTOR ACTION OF THE OTHER (AGONIST)

- **Two Drugs Produce Opposite Effects On A Same Physiological Function By Acting On Same Receptors**
- **Receptor Antagonism Can Be Competitive Or Non-Competitive**

Pharmacological Antagonism



Pharmacological Antagonism

Caffeine versus **Morphine**

Physiological Antagonism



Physiological Antagonism

Insulin versus **Glucagon**

Chemical Antagonism



Chemical Antagonism

Dimercaprol versus **Arsenic**

Adverse Drug Reactions

A Augmented

Side Effects

Upset Stomach



Side Effects

Upset Stomach

Dose-Dependent



Dose Effects

Dose-Dependent

B Bizarre

Allergy

Overdose, Organ Damage



Toxic Effects

Overdose, Organ Damage

Dose-Independent

C Chronic

Tolerance



Idiosyncratic Reactions

Dose-Independent

D Delayed

Teratogenicity

Birth Defects



Tolerance

Unpredictable & Rare

Dose-Dependent



Drug-Drug Interactions

Long-Term Use



Drug-Drug Interactions

Dose-Dependent

QUALITATIVE INTOLERANCE IDIOSYNCRASY

an unusual, individual response to certain drugs due to deficiency/absence of certain enzymes in a drug recipient

OR

genetically-controlled abnormal reaction to a drug

e.g. hemolysis produced by Primaquine and Sulfonamides in patients with G.6.P.D deficiency.

SPECIFIC ADVERSE REACTIONS

TERATOGENECITY

Capacity of a drug to cause foetal abnormalities when administered to the pregnant mother during 1st three months of pregnancy



Examples of teratogenic drugs:

- Phenytoin
- Valproic acid
- Captopril
- Warfarin



DRUG TOLERANCE

Reduction in the response due to continued use or repeated administration of drug

E.G of Drugs

TACHYPHYLAXIS (ACUTE TOLERANCE)

Rapid reduction in responsiveness due to repeated, frequent administration of drug

i.e. **Ephedrine, Tyramine, Amphetamine**

Pharmacodynamic Drug-Drug Interactions

Synergism

$$1 + 1 = 3$$



Potentiation

$$0 + 1 = 2$$



Addition

$$1 + 1 = 2$$



Antagonism

$$1 + 1 = 0$$



ADDITIVE EFFECT OR SUMMATION

Total pharmacological effect produced by the concomitant use of two or more drugs is equal to the sum of their individual effects:

$$1 + 1 = 2.$$

Example:

Combination of **Ephedrine** and **Theophylline** in the treatment of asthma.

SYNERGISM

Total pharmacological effect produced by concomitant use of two or more drugs is higher than the sum of their individual effects

$$1 + 1 \geq 2$$

Example:

- **Codeine + Aspirin: ↑ Analgesia.**

POTENTIATION

- Enhancement of the effect of one agent by another; so that the combined effect is more than the sum of their individual effects.
- **one agent has no effect when given alone but increases the effects of other co-administered drug.**

$$1 + 0 \geq 2$$

Example:

- **Levodopa + Carbidopa: > benefit in Parkinsonism**
- **Ampicillin + Clavulanic acid: > antibacterial effect**

ANTAGONISM

- the phenomenon of opposing effects when two or more drugs are given together

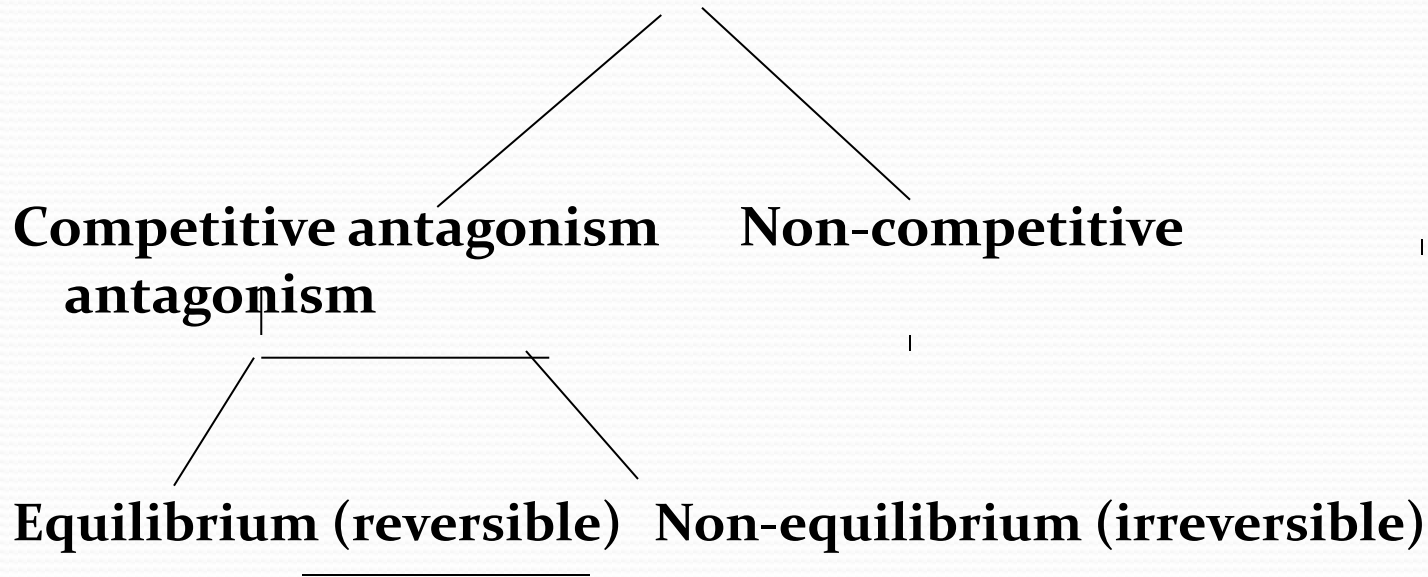
There are three types of antagonism

- **Chemical antagonism.**
- **Physiological antagonism.**
- **Pharmacological antagonism.**

PHARMACOLOGICAL ANTAGONISM

- two drugs produce opposite effects on a same physiological function by acting on same receptors

PHARMACOLOGICAL ANTAGONISM



Pharmacodynamic **Drug** Interactions

How Drugs Interact in the Body

Addition

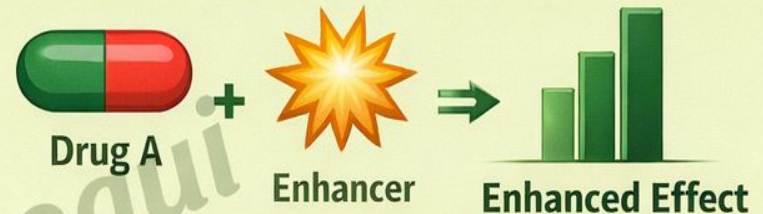
Drug A + Drug B = Combined Effect



$$1 + 1 = 2$$

Potentiation

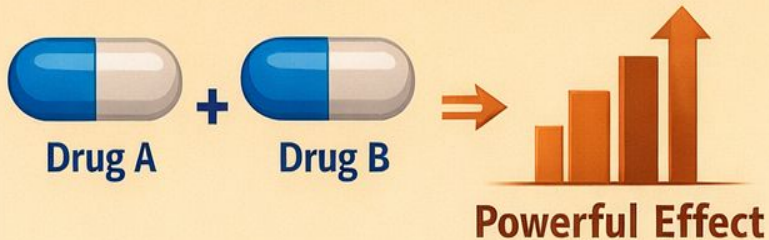
Drug A + Enhancer = Stronger Effect



$$1 + 0 = 3$$

Synergism

Drug A + Drug B = Greater Effect



$$1 + 1 = 5$$

Antagonism

Drug A + Drug B = *Reduced Effect*



$$1 + 1 = 0$$

THANK YOU